Dynamic programming

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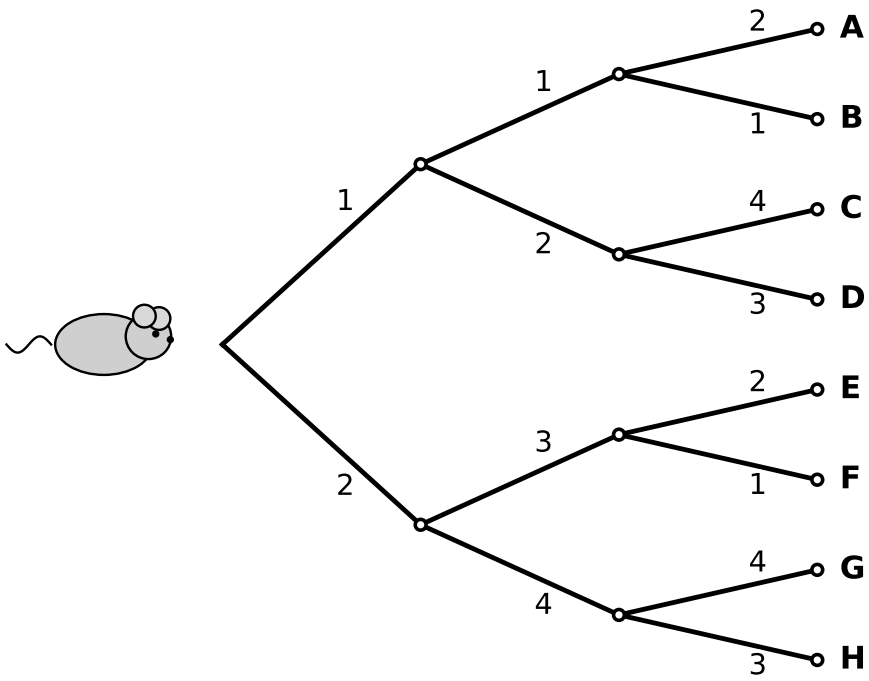
UM Math Circle, 2/19/2026

1 Mouse in a maze and dynamic programming

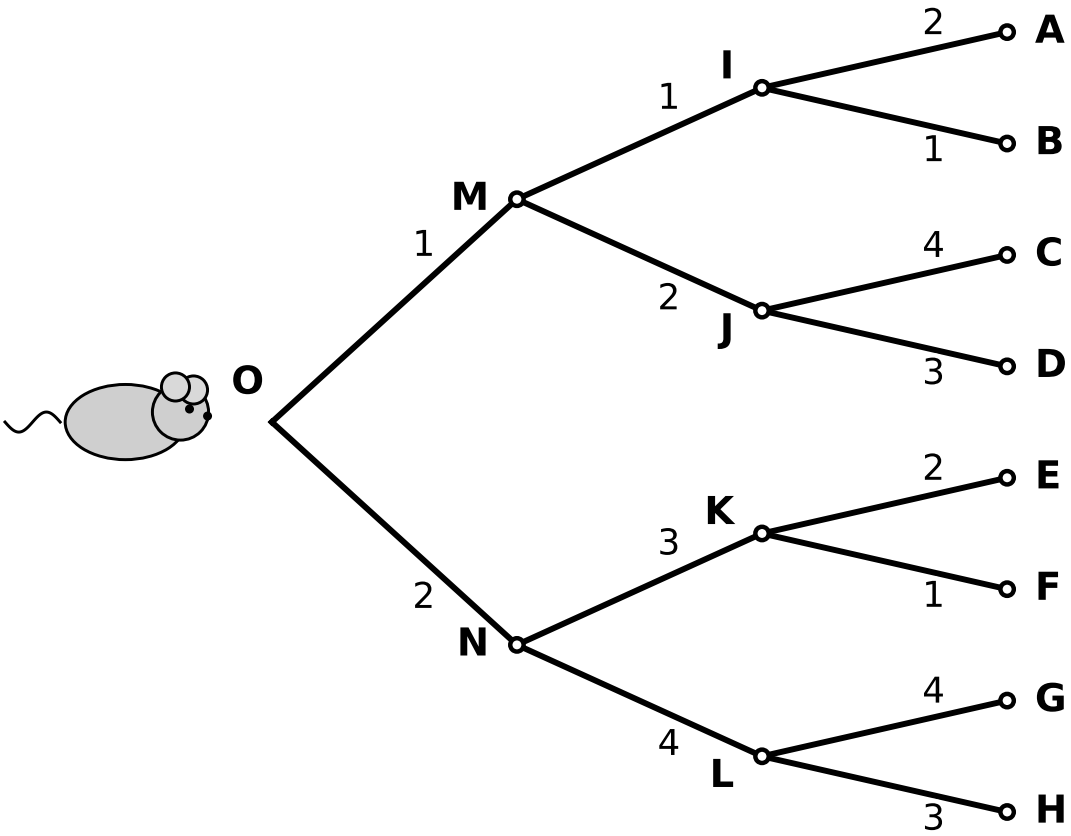
We can still do the maze problem by hand if the maze has 3 stages.
However, what if there are 4 stages? (Q: How many possible choices?)
How about 5 stages? (Q: how many possible choices?)

Let's introduce a new idea called "backward induction".
It is an important component of "dynamic programming".

- Not about coding.
- But it is a key mathematical technology used in "reinforcement learning" used in "machine learning".



Let's name the internal nodes.



<u>Bellman equation</u>

$$V(\mathrm{M}) = \underbrace{\text{max over all choices}}_{\text{go to I or J}} \{1 + V(\mathrm{I}), 2 + V(\mathrm{J})\}$$

In this example, $V(\mathrm{I}) = 2$ and $V(\mathrm{J}) = 4$, so

- What's $V(\mathrm{M})$?
- Which branch should the mouse select at location M?
- What's $V(\mathrm{N})$ and $V(\mathrm{O})$?
- Based on this, what's the best path starting location O?

The Bellman equation is used in

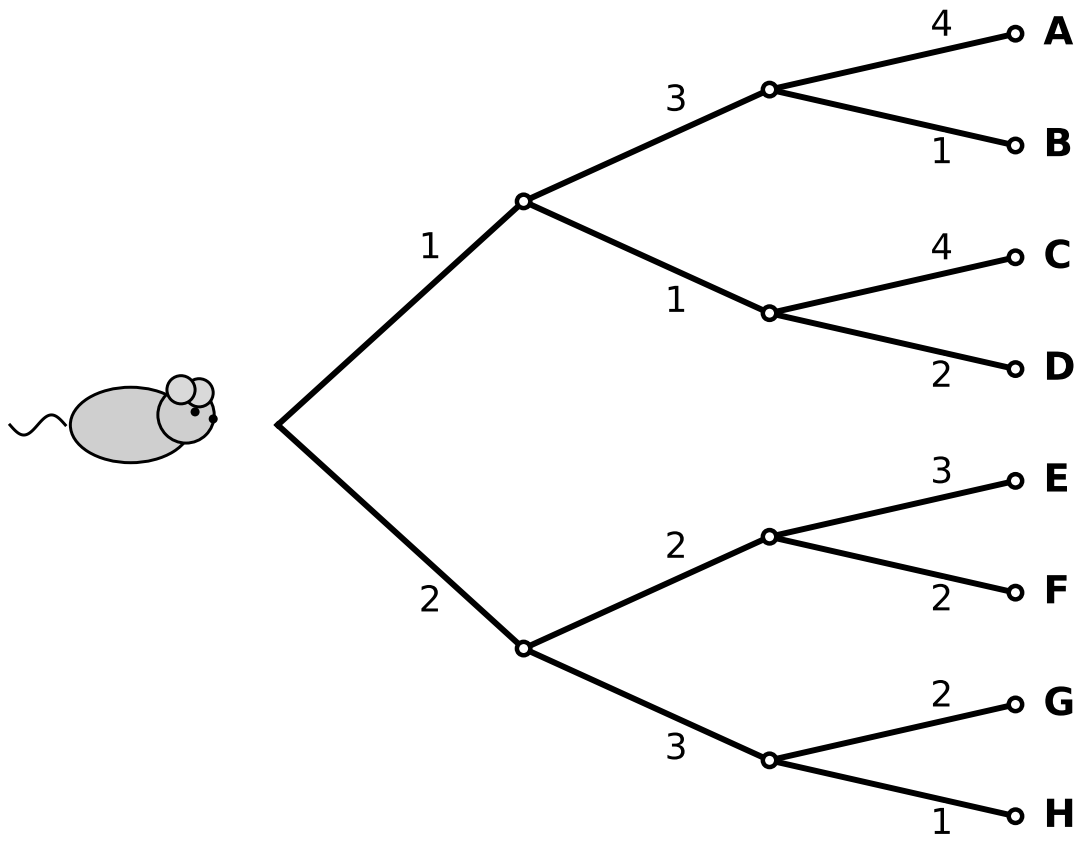
- robots, cars, finding best routes
- delivery companies (UPS, Fedex) to find fastest and cheapest ways
- banks findindg the best way to invest money over time
- hospital scheduling staff
- $\cdots$

How to do this?

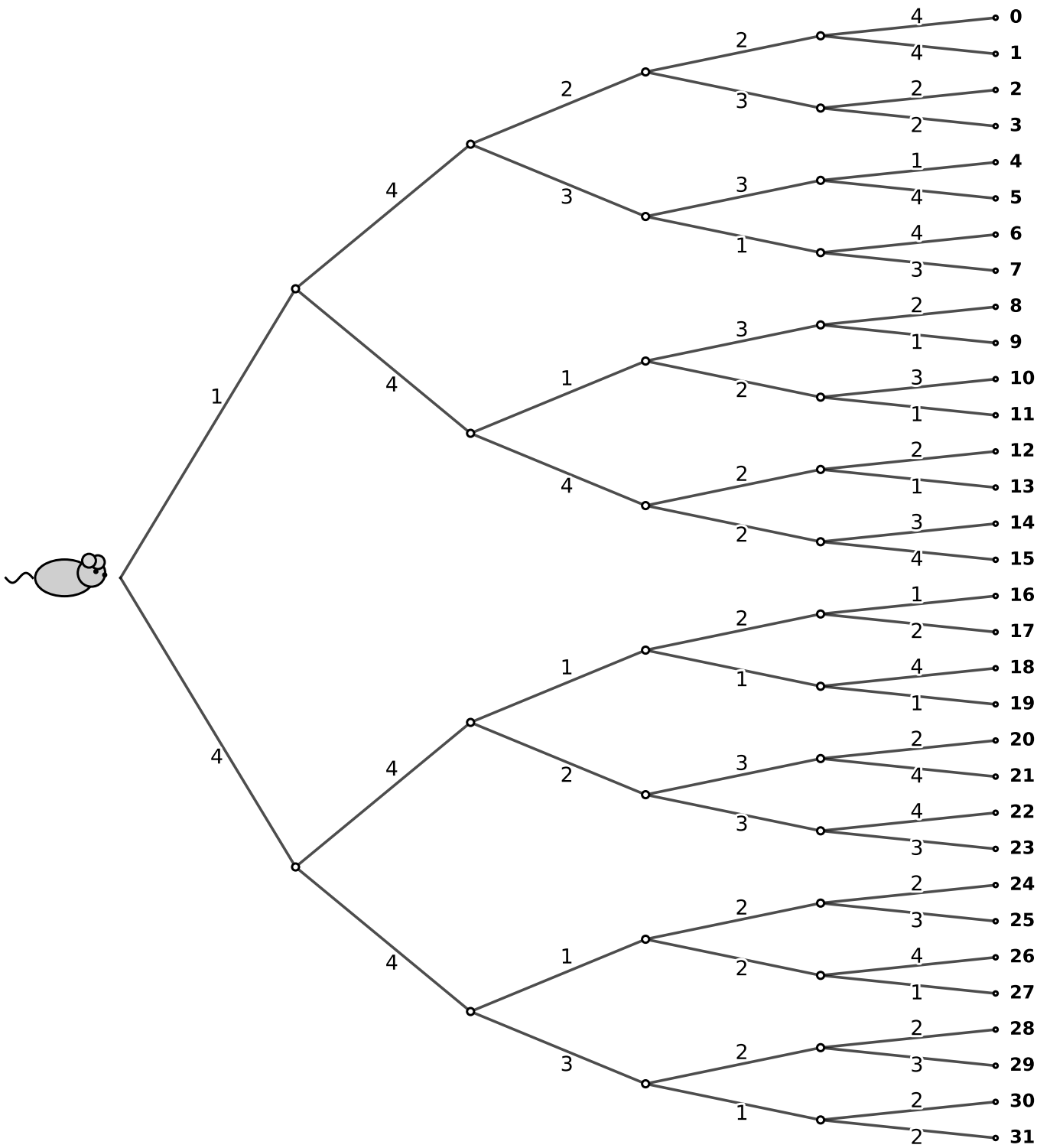
- Work backward.

- Determine the <u>value</u> of each <u>state</u> (= location)

Problem 1 In the following maze, get the mouse's optimal path and the total reward the mouse gets by taking the optimal path. Use backward induction.



Problem 2 How about the following maze? Get the optimal path and total reward.


1
4
4
4
2
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4
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3
1
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1
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4
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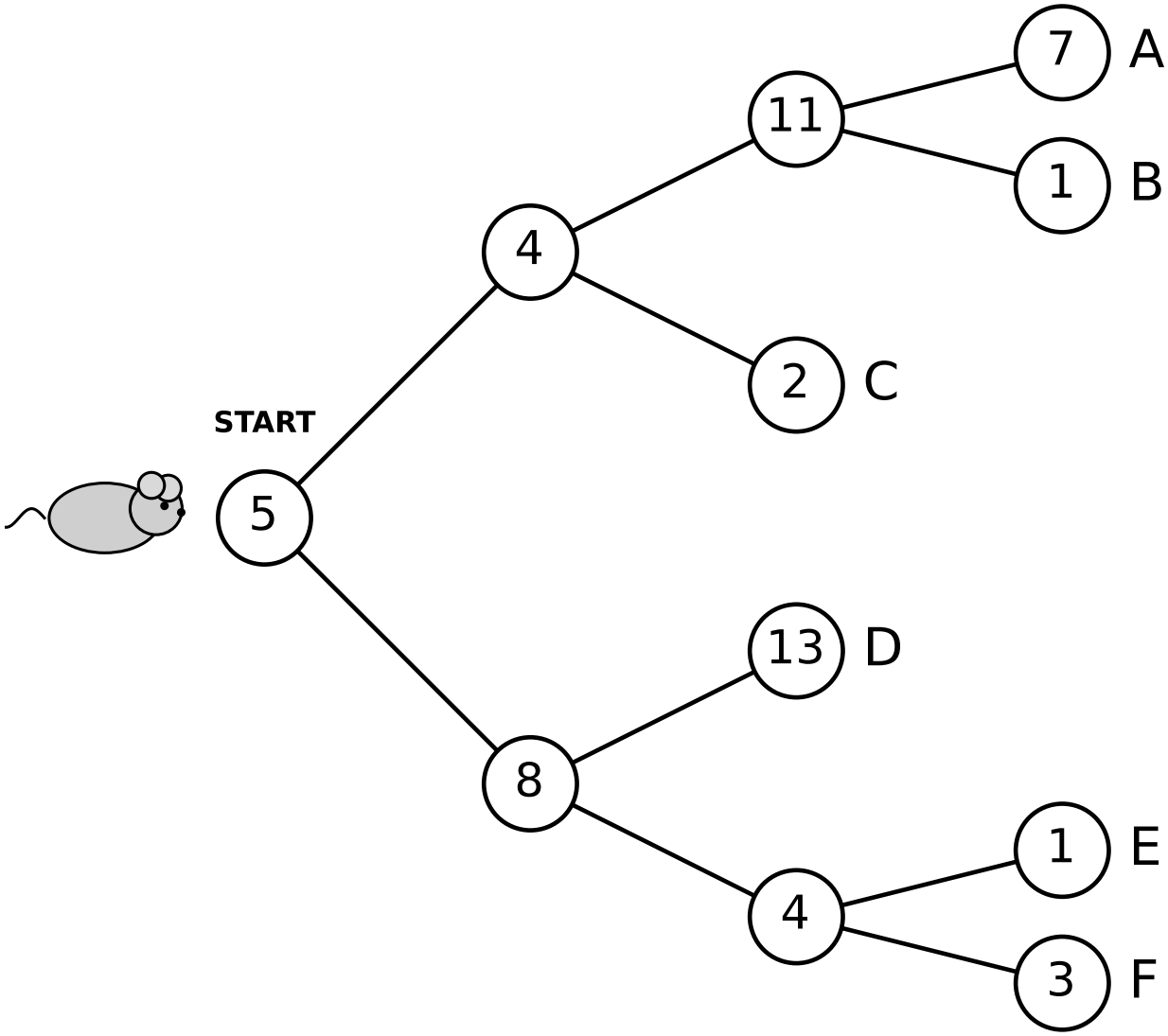
3
2
1
2
1
2
1

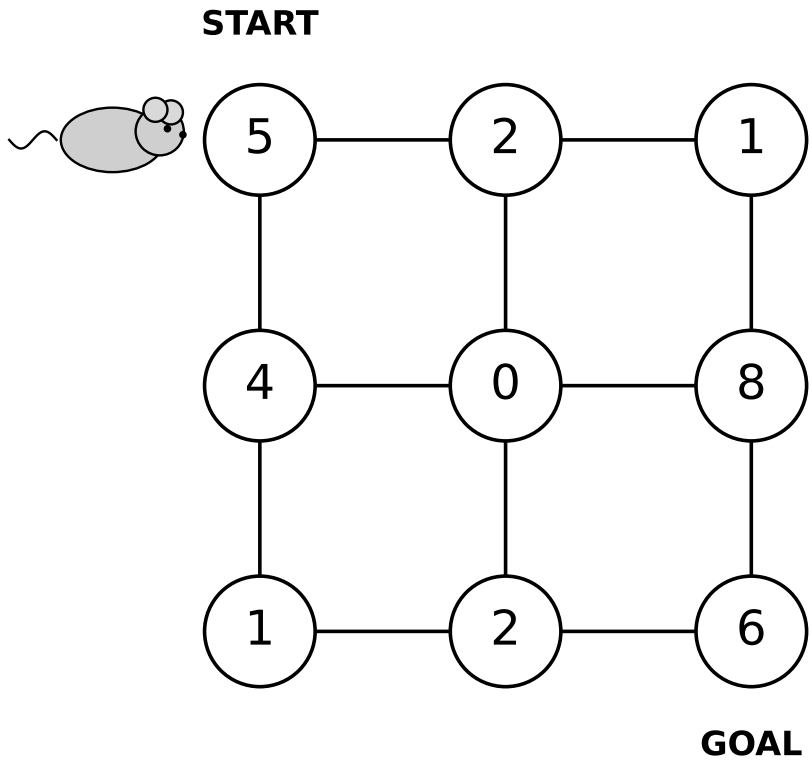
Problem 3 How about this uneven tree? Get the optimal path and total reward. A long path is not necessarily better.

Now the rewards are placed on nodes rather than branches, but it does not matter. Mice do not care!

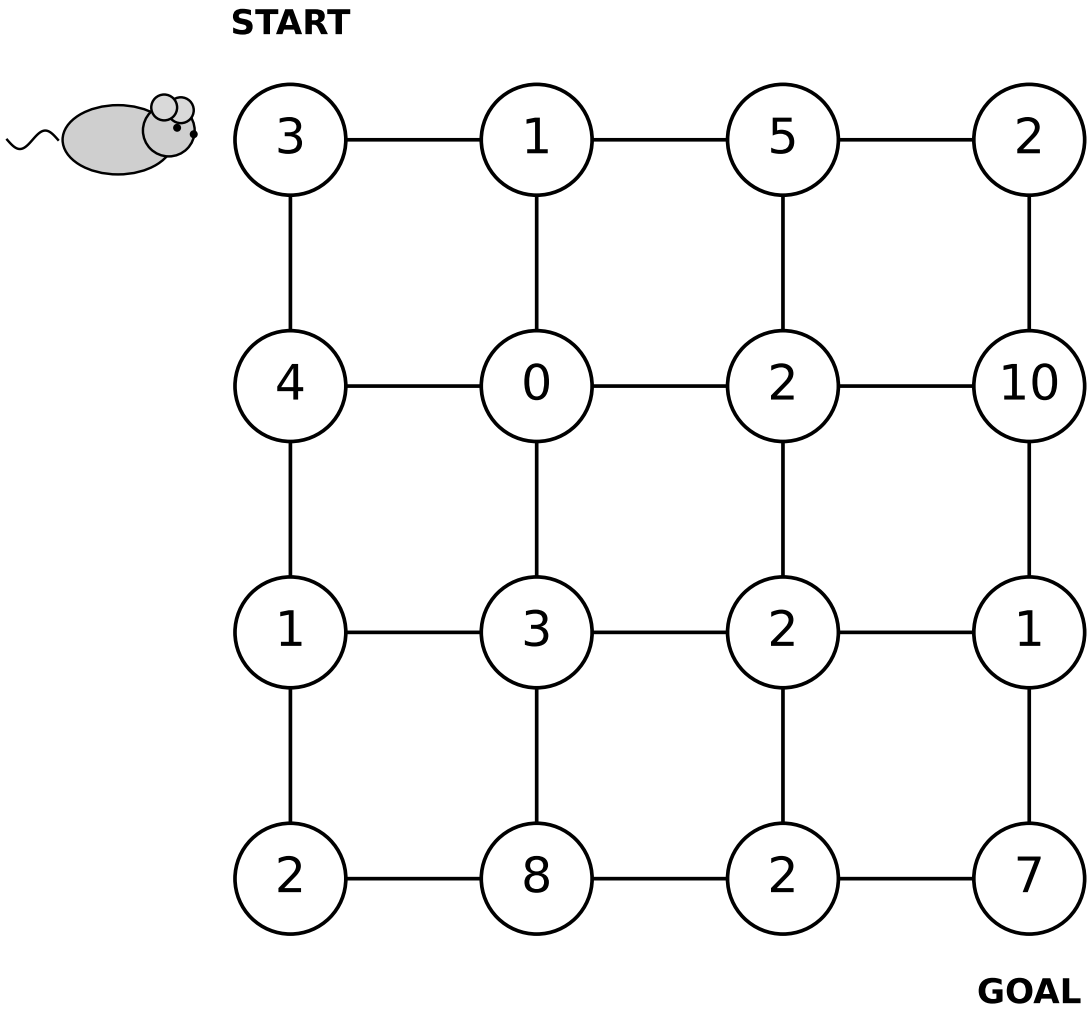
Start by setting $V(A) = 7$, $V(B) = 1$, $V(C) = 2$, and so forth. As usual, work backward (from right to left of the figure).



Problem 4 How about the following square maze? The mouse is only allowed to go RIGHT or DOWN.

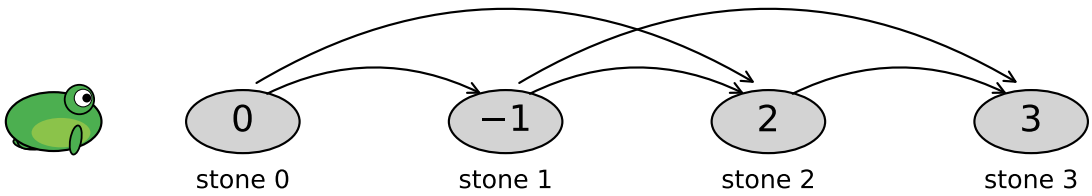


Problem 5 How about the following bigger square maze? The mouse can only go RIGHT or DOWN.



2 Jumpy frog path problem

There are stepping stones in a river, and there is a frog.



A frog is on stone 0 and wants to go to stone 3. In each jump, the frog can either go to the next stone or skip one stone. On each stone, there is a reward (negative value meaning punishment) the frog can get (food, poison, predation risk, etc.). What's the best path the frog should take?

Let's do this one together:

Q: How many possible paths?

1. Write the Bellman equation:

$$V(k) = (\text{reward at stone } k) + \max\{V(k+1), V(k+2)\}.$$

2. Figure out what we know initially:

   Base case: $V(3) = \text{reward at stone } 3 = 3$.

3. Do "backward induction" using the Bellman equation:

$$V(2) = (\text{reward at stone } 2) + V(3) = 2 + 3 = 5.$$

   (Note that the frog does not have a choice at stone 2.)

$$V(1) = (\text{reward at stone } 1) + \max\{V(2), V(3)\} = -1 + \max\{5, 3\} = 4,$$
$$V(0) = (\text{reward at stone } 0) + \max\{V(1), V(2)\} = 0 + \max\{4, 5\} = 5.$$

4. Say the solution:

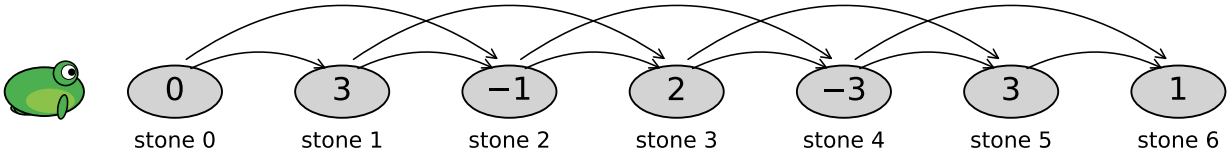
   The optimal path is $0 \to 2 \to 3$.
   The total reward the frog gets is $V(0)$, which is 5.

Problem 6 How about the following step stone sequence? Optimal path and total reward?

The Bellman equation for this game is:

$$V(k) = \begin{cases} R(k) + \max\{V(k+1), V(k+2)\} & \text{if } k = 0, \ldots, 4 \text{ (the frog has two choices)}, \\ R(k) + \max\{V(k+1)\} & \text{if } k = 5, 6 \text{ (the frog has just one choice)}. \end{cases}$$

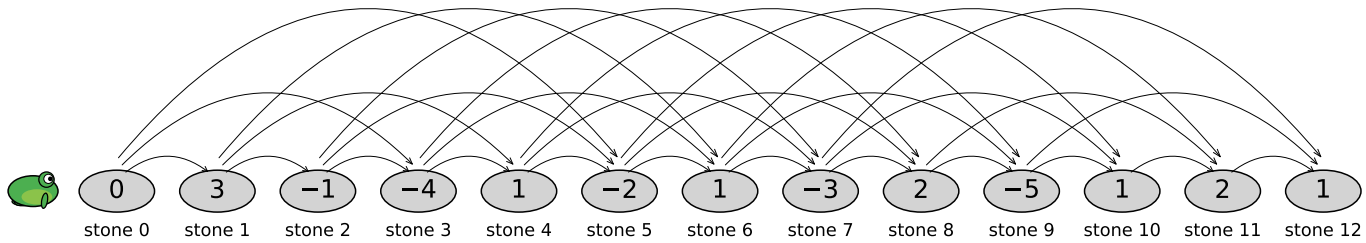
$R(k)$ means the reward received at stone k.



Problem7 How about the following case? The frog can now go to the next stone, skip two stones, or skip four stones.

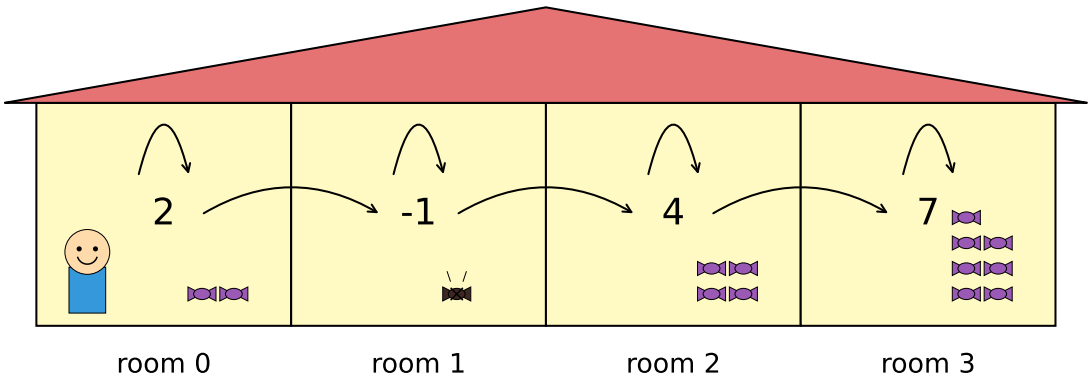
Now the Bellman equation is

$$V(k) = \begin{cases} R(k) + \max\{V(k+1), V(k+3), V(k+5)\} & \text{if } k = 0, \ldots, 7 \text{ (the frog has three choices)}, \\ R(k) + \max\{V(k+1), V(k+3)\} & \text{if } k = 8, 9 \text{ (the frog has two choices)}, \\ R(k) + \max\{V(k+1)\} & \text{if } k = 10, 11, 12 \text{ (the frog has just one choice)}. \end{cases}$$



3 Discounted candy collection

You are in room 0.



- You can either move RIGHT (to the next room) or STAY each day.
- Each room has a candy dispenser of a different reward.
- It may dispense a bitter candy! (See room 1).
- It dispenses the known number of candies (2, -1, 4, or 7 depending on the room) each day.

Q: What would you do?

However, for me, what I get now is more important than what I get in the future.

- This is called "discounted utility". Appearing in economics theory, experimental psychology, etc.

So, what you want to maximize is:

$$\text{reward}(t) + \frac{1}{2}\text{reward}(t+1) + \frac{1}{4}\text{reward}(t+2) + \frac{1}{8}\text{reward}(t+3) + \cdots . \tag{1}$$

t is today, $t+1$ is tomorrow, $t+2$ is the day after tomorrow, etc.

You can reach room 3 in three days, but the seven candies you can get after 3 days give you a $\frac{1}{8} \times 7$ value only. Should you go there or stay in room 0?

Let's do this problem together.

1. Write down the Bellman equation

$$V(s) = R(s) + \max\left\{\frac{1}{2}V(s), \frac{1}{2}V(s+1)\right\}. \tag{2}$$

   - $V(s)$ is the value of the room s.
   - The first option is STAY, and the second option is RIGHT.
   - $R(s)$ is the reward you get today in room s (then you decide to STAY or go RIGHT).

2. Figure out what we know initially:

   Nothing we can do. We do not know even $V(3)$ yet.

3. Do "backward induction" using the Bellman equation:

$$V(3) = R(3) = \frac{1}{2}V(3) = 7 + \frac{1}{2}V(3). \tag{3}$$

   (You can only STAY if you are in room 3.)

   Q: $V(3)$?

   Today, you get 7, tomorrow's candy is counted as 3.5, the day after tomorrow's is 1.75, $\ldots \to$ total is 14.

$$V(2) = 4 + \max\left\{\frac{1}{2}V(2), \frac{1}{2}V(3)\right\}.$$

   If STAY in room 2 is better, then $V(2) = 4 + \frac{1}{2}V(2)$, which leads to $V(2) = 8$.
   If (going) RIGHT in room 2 is better, then $V(2) = 4 + \frac{1}{2}V(3) = 4 + \frac{1}{2} \times 14 = 11$.
   So, better to go RIGHT if you are in room 2, and $V(2) = 11$.

$$V(1) = -1 + \max\left\{\frac{1}{2}V(1), \frac{1}{2}V(2)\right\}.$$

   If STAY in room 1 is better, then $V(1) = -1 + \frac{1}{2}V(1)$, which leads to $V(1) = -2$. This sounds really bad.
   If (going) RIGHT in room 1 is better, then $V(1) = -1 + \frac{1}{2}V(2) = \frac{9}{2}$.
   So, better to go RIGHT if you are in room 1, and $V(1) = \frac{9}{2}$.

$$V(0) = 2 + \max\left\{\frac{1}{2}V(0), \frac{1}{2}V(1)\right\}.$$

   If STAY in room 0 is better, then $V(0) = 2 + \frac{1}{2}V(0)$, which leads to $V(0) = 4$.
   If (going) RIGHT in room 0 is better, then $V(0) = 2 + \frac{1}{2}V(1) = \frac{17}{4}$, which is larger than 4.
   So, better to go RIGHT if you are in room 0, and $V(0) = \frac{17}{4}$.

4. Say the solution:

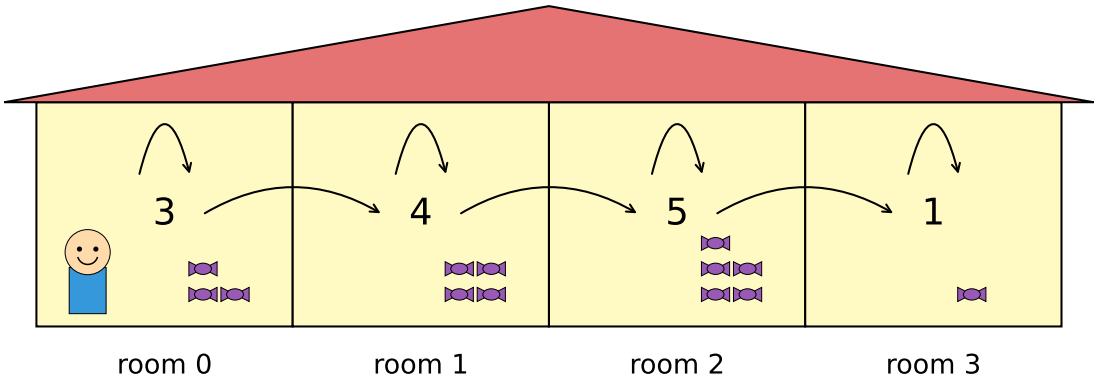
   The optimal behavior is go RIGHT RIGHT RIGHT toward room 3.

Problem 8 With a "discount factor" $r = \frac{1}{2}$, the best solution was to go RIGHT RIGHT RIGHT.

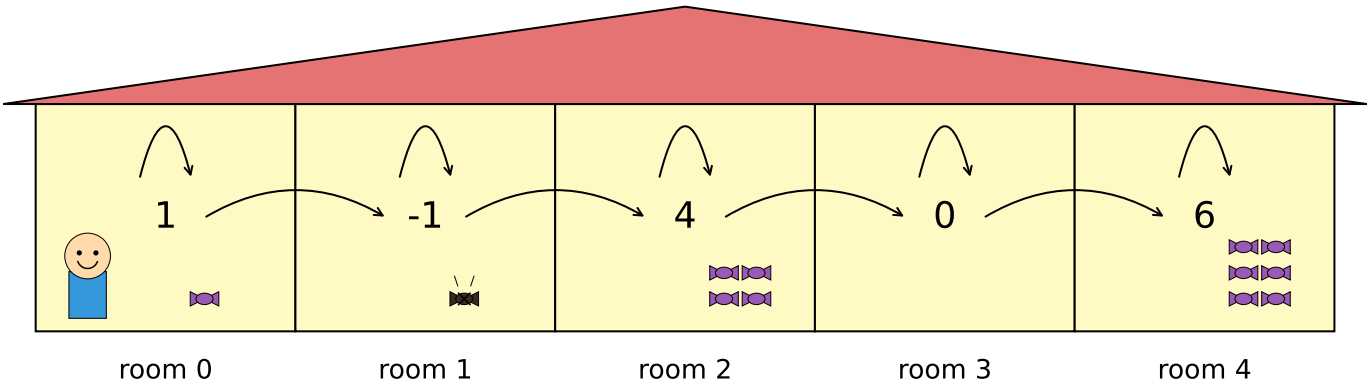
What if you *really* want to the candy now than later?

Change the discount factor from $r = \frac{1}{2}$ to $r = \frac{1}{8}$, and get $V(3)$, $V(2)$, $V(1)$, and $V(0)$ to figure out the optimal behavior.

Problem 9 What is the optimal behavior in the following case with $r = \frac{1}{2}$?



Problem 10 What is the optimal behavior in the following case with $r = \frac{1}{2}$?
How about $r = \frac{3}{4}$ (You care a lot about the future)?



4 Edit distance

The (minimum) edit distance is the number of (i) insertions of a single letter, (ii) deletions of a single letter, or (iii) substitutions of a single letter to turn one word to another?

Q: What is the edit distance between

1. "CART" and "CAR"?
2. "FROM" and "FORM"?
3. "NEAR" and "EARN"?
4. "SUNDAY" and "SATURDAY"

$d(i, j)$ = minimal number of edits to turn the first i letters of "NEAR" into the first j letters of "EARN".

- $d(1, 1) =$ "N" $\rightarrow$ "E" $= 1$,
- $d(1, 2) =$ "N" $\rightarrow$ "EA" $= 2$.

Bellman equation:

$$d(i, j) = \begin{cases} 1 + \min\{d(i-1, j), d(i, j-1), d(i-1, j-1)\} & \text{if } S(i) \neq T(j), \\ d(i-1, j-1) & \text{if } S(i) = T(j). \end{cases} \tag{4}$$

We do it via a <u>d-table</u>.

Problem 11

1. What is the edit distance from "NEAR" to "EARN"? We know the answer more intuitively, but check it by creating the d-table.
2. From "LEAST" to "STEAL"?
3. From "CAMELY" to "BALLET"?
4. From "SELLER" to "REELLS"?
5. From "SUNDAY" to "SATURDAY"?